

Robust Face Anti-Spoofing under Lighting Variation Using Deep Pixel-wise Supervision on Lightweight CNNs

David Christian Golden Mahaviro
Computer Science Department
School of Computer Science (SOCS)
Bina Nusantara University (BINUS)
Jakarta, Indonesia
david.mahaviro@binus.ac.id

Edwin Antonie
Computer Science Department
School of Computer Science (SOCS)
Bina Nusantara University (BINUS)
Jakarta, Indonesia
edwin.antonie@binus.ac.id

Hasan
Computer Science Department
School of Computer Science (SOCS)
Bina Nusantara University (BINUS)
Jakarta, Indonesia
hasan002@binus.ac.id

Irene Anindaputri Iswanto
Computer Science Department
School of Computer Science (SOCS)
Bina Nusantara University (BINUS)
Jakarta, Indonesia
irene.iswanto@binus.edu

Arya Krisna Putra
Computer Science Department
School of Computer Science (SOCS)
Bina Nusantara University (BINUS)
Jakarta, Indonesia
arya.putra006@binus.ac.id

Abstract—Facial recognition is a central piece of modern digital security. However, these systems are often vulnerable to presentation attacks like printed photos or video replays, with this vulnerability worsening by extreme lighting. This study proposes a two-stage real-time authentication framework to solve this problem. The first stage utilizes Deep Pixel-wise Binary Supervision (DeepPixBis) to detect spoofing attempts by examining local texture representations pixel by pixel. ArcFace architecture was used for the real identity verification in the second stage. Three lightweight CNN backbones (ShuffleNetV2, MobileNetV2 and EfficientNet-B0) were evaluated to find out which architecture fits best for resource-constrained edge devices. Testing on the challenging Protocol 4 of OULU-NPU dataset also showed that EfficientNet-B0 achieved better results than the others. It had an Attack Presentation Classification Error Rate (APCER) of 3%, a Bona Fide Presentation Classification Error Rate (BPCER) of 27%, and an Average Classification Error Rate (ACER) of 15%. Optimal model integrated into a live webcam based system. To conclude, results demonstrate that DeepPixBis is a very efficient and effective solution for robust face anti-forgery in unpredictable illumination conditions.

Keywords—Face Anti-Spoofing, DeepPixBis, Deep Pixel-wise Binary Supervision, Presentation Attack Detection, Lightweight CNN, Edge Computing, OULU-NPU Dataset.

I. INTRODUCTION

Face recognition is one of the most popular biometric authentication methods in today's digital systems such as smartphones, laptops and access control devices. With the development of deep learning techniques, face recognition systems have achieved great improvements in accuracy and reliability [23]. A notable approach is proposed in FaceNet [2] that learns discriminative facial embeddings that enable accurate face identification and verification. Furthermore, later models such as ArcFace [4] have improved the angular margin separation significantly, which allows large scale deployment. However, face recognition systems are still vulnerable to presentation attacks, which are also known as face spoofing, where an attacker tries to fool the system with printed photos, replayed videos, or images shown on digital screens [5], [14].

To solve this problem and to distinguish the real faces from spoofing attacks, a number of Face Anti-Spoofing (FAS) techniques have been proposed by researchers. These techniques are often tested on benchmark

datasets such as OULU-NPU [1], CelebA-Spoof [12], and CASIA-SURF [25], which cover different attack scenarios and environmental settings relevant to face authentication systems on mobile devices. In controlled laboratory environments, numerous FAS methods, spanning from handcrafted texture-based techniques to deep learning methods, have demonstrated promising outcomes. However, they generally perform poorly in real world settings [22].

The lighting variation is one of the main challenges for the performance of face anti-spoofing systems. The changes in the intensity of illumination, its direction and the distribution of shadows [15] can significantly influence the visual appearance of facial images captured by cameras. Such variations can lead to inconsistencies in extracted features, making it more difficult for models to correctly distinguish between real faces and spoofing artifacts [29]. Thus, many existing methods are challenged to perform robustly under varying illumination [10], [26].

Therefore, this paper suggests a robust facial anti-spoofing framework against difficult lighting variations by using Deep Pixel-wise Binary Supervision (DeepPixBis). DeepPixBis utilizes a novel pixel-by-pixel supervision scheme, different from the usual binary classification, so the model can learn local texture patterns related to the spoofing, thus improving its ability to distinguish the genuine presentation and the attack. Three lightweight CNN backbones are studied in the DeepPixBis framework for evaluation of their potential for utilization on resource-limited devices, namely ShuffleNetV2, MobileNetV2 and EfficientNet-B0. We further incorporate the proposed system into a real-time webcam-based authentication pipeline using ArcFace for identity verification, providing an efficient and practical solution for biometric security of edge devices.

II. RELATED WORKS

A. Face Recognition and Spoofing Vulnerabilities

Face recognition systems, such as FaceNet proposed by Schroff et al. [2], have achieved high accuracy by learning discriminative facial embeddings. However, these systems are still highly vulnerable to presentation attacks with printed photos or digital screens [5]. The initial Face Anti-Spoofing (FAS) methods were based on texture

analysis [6] and eye-blinking detection [15], but usually suffer from significant performance degradation when applied to real-world scenarios with uncontrolled lighting conditions. Therefore, recent works have moved to deep learning paradigms with auxiliary supervision [7], deep pixel-wise binary supervision [9] and disentangled feature learning [24] to disentangle spoofing traces from genuine facial representations [21].

B. Robustness to Lighting Variations in FAS

Illumination variability in real-world environments is one of the hardest challenges for FAS systems [19]. The changing intensity, direction, and shadow patterns of light change the appearance of a face and spoofing artifacts drastically, which makes it hard for the model to differentiate between them. To solve this, Boulkenafet et al. [1] released the OULU-NPU dataset which has a Protocol 4 to evaluate the robustness of a model under extreme and diverse lighting conditions. Although methods such as 3D Central Difference Convolutional Networks [8], [13] and dynamic grayscale snippet analysis [30] have demonstrated outstanding results, the majority of existing methods are unable to robustly perform under such conditions without resorting to computationally expensive architectures, domain generalization techniques [11], [18], or specialised multi-sensor hardware.

C. Lightweight CNN Architectures for Edge Computing

Deployment of Face Anti-Spoofing (FAS) systems on resource-constrained devices like regular laptops requires architectures that balance detection performance with computational efficiency [27]. Several lightweight Convolutional Neural Network (CNN) architectures have been proposed to fulfill this requirement. Sandler et al proposed MobileNetV2 [3], which adopts inverted residual blocks and linear bottlenecks to reduce the computational cost and maintain effective feature representation. Zhang et al. [27] proposed ShuffleNetV2, which improves the efficiency by channel shuffling and point-wise group convolutions and is suitable for applications with limited hardware resources.

More recently, Tan and Le [31] proposed EfficientNet, which scales up network depth, width and input resolution using a compound scaling method. Its baseline variant EfficientNet-B0 provides a good trade-off between model capacity and computation demand. Although these architectures have achieved good results in generic image classification tasks, their effectiveness for Face Anti-Spoofing in challenging illumination conditions is still not sufficiently investigated. This is even more the case for edge-device deployments where the anti-spoofing module is the first security layer before identity verification. In this study, a comparative evaluation of MobileNetV2, ShuffleNetV2 and EfficientNet-B0 will be conducted to determine the best lightweight architecture for pixel-wise supervised FAS systems in client-specific anomaly detection settings [16].

D. Deep Pixel-wise Binary Supervision for Face Anti-Spoofing

In recent years, Face Anti-Spoofing (FAS) has evolved from the conventional binary classification paradigm to pixel-wise supervision methods. A popular framework is Deep Pixel-wise Binary Supervision (DeepPixBis) by George and Marcel [9]. Unlike conventional CNN-based classifiers that produce a single global prediction for an input image, DeepPixBis produces pixel-level spatial maps that provide probabilities of authenticity or fakeness for local facial regions.

The core idea of DeepPixBis is to guide the network to learn localized texture patterns associated with fakeness rather than just global facial appearance. This method makes synthetic depth maps no longer needed in previous additional supervision methods and still has a strong detection ability. Furthermore, the pixel-level supervision mechanism improves model interpretability by allowing the visualization of regions that influence the final decision on authenticity.

Experimental results reported by George and Marcel [9] suggest that DeepPixBis achieves competitive performance on benchmark datasets like OULU-NPU and Replay-Mobile with only frame-level information. Those properties make DeepPixBis suitable for real-time authentication systems and for deployment on resource-constrained devices. Therefore, this study uses DeepPixBis as the main facial anti-spoofing framework and evaluates its performance with different lightweight CNN architectures under challenging lighting conditions.

III. METHODOLOGY

The proposed Face Anti-Spoofing (FAS) system is assessed by an experimental study with the benchmarking approach in this very research. The system implemented is a two-stage real-time authentication pipeline. The first stage operates as a liveness detection gate based on Deep Pixel-wise Binary Supervision (DeepPixBis) combined with light-weight Convolutional Neural Network (CNN) backbones. The second stage is for facial recognition for identity verification.

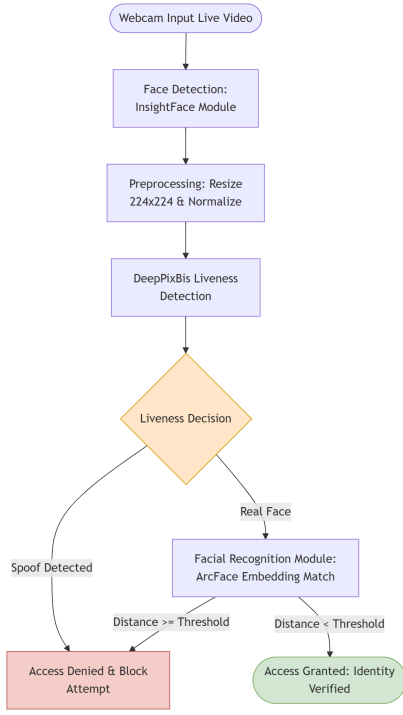


Fig. 1. Proposed Pipeline of the Face Anti-Spoofing and facial recognition

A. Dataset

The main dataset used is the OULU-NPU mobile face presentation attack dataset [2], mainly focusing on Protocol 4 which is to test the robustness of the model against extreme lighting variations. The dataset is strictly divided into Train, Development and Test sets.

B. Data Preprocessing

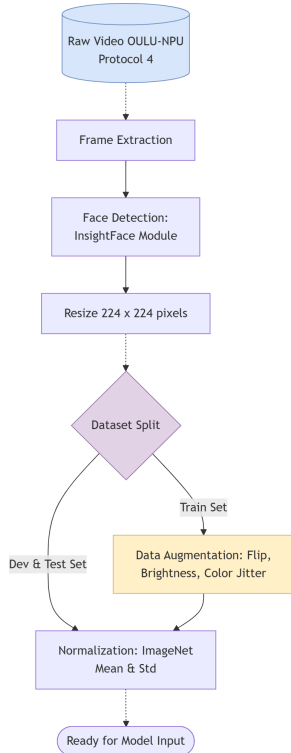


Fig. 2. Proposed data preprocessing pipeline for the face anti-spoofing model.

Once the dataset is chosen, it is important to set up a suitable processing pipeline so that the model can learn effectively from the data. First, we implement face detection on the raw video frames using the state-of-the-art InsightFace module, that accurately detects the facial bounding boxes. The cropped facial regions are then resized to 224×224 pixels to meet the input requirement of the selected lightweight networks.

For the training set, random horizontal flipping, brightness jitter and color jitter are used for data augmentation to simulate variations in illumination. Finally, all image splits are normalized with standard ImageNet mean values ([0.485, 0.456, 0.406]).

C. Liveness Detection Architecture

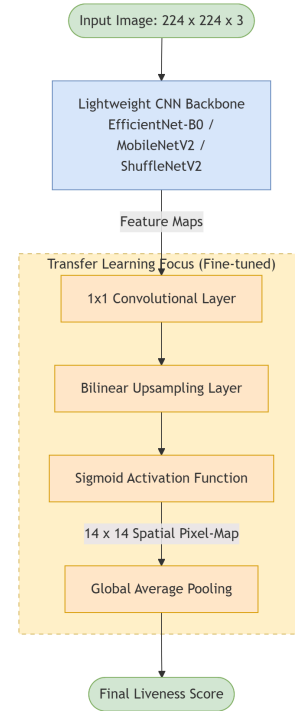


Fig. 3. Proposed Liveness Detection Architecture

For the face anti-spoofing (FAS) task, the system shifts from the traditional global image classification to Deep Pixel-wise Binary Supervision (DeepPixBis) [9]. To achieve computational efficiency suitable for edge devices, three lightweight architectures pre-trained on ImageNet are benchmarked as feature extraction backbones: EfficientNet-B0, MobileNetV2, and ShuffleNetV2.

To adapt the networks for pixel-wise supervision, the original fully connected classification heads are removed. Instead, the extracted feature maps are passed through a 1x1 convolutional layer, followed by a bilinear upsampling layer and a Sigmoid activation function to generate a 14 x 14 spatial pixel-map. Each pixel in this map represents the localized probability of a genuine or spoof presentation. The final global liveness score is obtained by calculating the mean of this 14 x 14 spatial map.

D. Training Protocol

The models are trained utilizing a Transfer Learning strategy to optimize computational resources. The base weights of the backbones are frozen, and only the newly

added pixel-wise convolutional head and upsampling layers (comprising approximately 21,000 parameters) are fine-tuned.

The network is optimized using the Adam optimizer with a learning rate of 1×10^{-4} . The training was performed for 15 epochs with the metric to evaluate the pixel-wise predictions being the Binary Cross-Entropy (BCE) Loss function. The benefit of this method is that it prevents overfitting and there is no overhead of high computation. The models converge quickly and learn local texture features instead of memorizing the training distribution.

E. Evaluation Protocol

As the main metric for the final evaluation on the Test set, the Average Classification Error Rate (ACER) is adopted. The metric is the average of the Attack Presentation Classification Error Rate (APCER) and the Bona Fide Presentation Classification Error Rate (BPCER) according to the common evaluation protocols in face anti-spoofing [2].

F. Facial Recognition Integration

The face recognition module receives input with faces that passed the liveness detection stage. At this stage the system focuses on identifying the individual by extracting distinctive facial features from the provided image. This transition ensures that only the real facial inputs are processed further, thereby increasing the overall reliability and security of the system.

The ArcFace model [5] from the InsightFace library is used to extract face embeddings from the system. The embeddings are then compared with a pre-registered local database using the cosine distance metric. A predetermined threshold (e.g. 0.75) is used to determine whether the detected face is a match for an authorized user. This approach is built on the embedding-based representation presented in FaceNet [1] to achieve accurate and efficient face recognition.

IV. RESULT AND DISCUSSION

This chapter provides experimental results and analysis of the proposed two-stage Face Anti-Spoofing (FAS) and facial recognition system. The evaluation is addressed to the implementation of Deep Pixel-wise Binary Supervision on three different lightweight architectures: EfficientNet-B0, MobileNetV2 and ShuffleNetV2.

The analysis is split into the main stages: training performance evaluation between different architectures, and models comparison with the OULU-NPU dataset based on the standard biometric error metrics ISO/IEC 30107-3: Attack Presentation Classification Error Rate (APCER), Bona Fide Presentation Classification Error Rate (BPCER), and Average Classification Error Rate (ACER).

This was followed by demonstration of a real time implementation of the integrated webcam based system. Moreover, the chapter explores the process of model selection by taking into account technical aspects, computational efficiency and overall performance to

motivate the most suitable architecture for deployment on edge devices.

A. Cross-Architecture Training Performance

The training convergence of the selected lightweight architectures was directly analyzed to evaluate their learning capabilities. The primary objective of this phase was to evaluate how well each model corresponds to the Deep Pixel-wise Binary Supervision method, which requires the network to extract local texture features and generate a 14x14 pixel-wise prediction map. One of the key elements which affect this adaptability is the representation capacity of the networks measured by the number of parameters. The architectural complexities of the models evaluated are presented in Table 1.

TABLE I. Parameter Complexity of the Evaluated Architectures

Architecture	Total Param.	Trainable Param.
ShuffleNetV2	1,274,430	20,826
MobileNetV2	2,244,954	21,082
EfficientNet-B0	4,028,630	21,082

All three models are lightweight and designed for edge deployment as shown in Table 1. EfficientNet-B0 has the highest representational capacity with around 4.02 million total parameters. On the other hand, the ShuffleNetV2 is the most constrained architecture. Specifically, the number of trainable parameters was approximately kept the same in all models by freezing the backbone layers and fine-tuning only the customized pixel-wise convolutional heads. Training loss trajectories were also plotted to observe these underlying architectural differences affect the learning process.

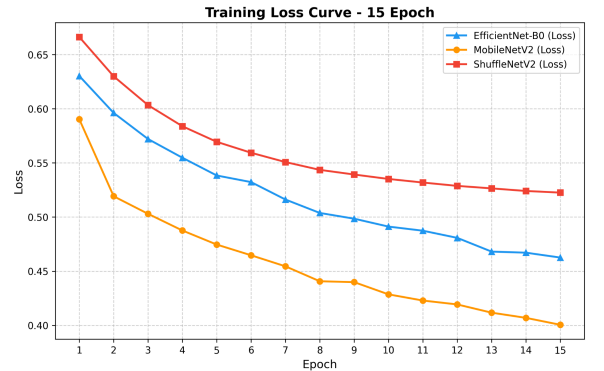


Fig. 4. Training loss trajectories of EfficientNet-B0, MobileNetV2, and ShuffleNetV2 over 15 epochs utilizing Deep Pixel-wise Binary Supervision.

The convergence behavior of the three architectures is shown in Fig. 4, which is directly related to their parameter capacities. The highest loss trajectory was found in the training phase of ShuffleNetV2. Its limited number of parameters was insufficient to represent the complex material textures needed by the pixel-wise loss function. On the other hand, MobileNetV2 showed an aggressive rate of convergence and attained the lowest numerical loss by the end of the 15th epoch. However, given the limited capacity

of the model, this very steep drop-off is a strong indicator of early model overfitting, where the network learned the training distribution instead of learning the presentation attack features for generalization.

On the other hand, EfficientNet-B0 showed the most optimal and balanced learning behavior. The training loss showed a smooth and steady decrease in architecture. In contrast to the aggressive drop of MobileNetV2, EfficientNet-B0 showed a healthy convergence trajectory, indicating that its larger, compound-scaled parameters were able to learn the underlying texture disparities between bona fide and spoofing artifacts without severely overfitting the training data.

B. Face Anti-Spoofing Benchmark on OULU-NPU

TABLE II. ISO/IEC 30107-3 Evaluation Metrics on OULU-NPU Protocol 4

Architecture	BPCER	APCER	ACER
ShuffleNetV2	100%	0%	50%
MobileNetV2	90%	0%	$\pm 45\%$
EfficientNet-B0	27%	3%	$\pm 15\%$

After the training phase, the models were tested against the unseen testing distribution of the OULU-NPU dataset under Protocol 4 which introduces extreme variations in ambient illumination and camera quality. The performance of the system was evaluated using standard ISO/IEC 30107-3 biometric metrics: BPCER, APCER and ACER. The evaluation results reveal a significant difference in how the various architectures dealt with the pixel-wise supervision task in real-world circumstances.

As shown in Table II, both ShuffleNetV2 and MobileNetV2 suffered from a major mode collapse during testing. ShuffleNetV2 reached a BPCER of 100% and an APCER of 0%, leading to an ACER of precisely 50%. Similarly, MobileNetV2 showed a very unbalanced performance with a BPCER above 90% and an APCER of 0%. These results show that both of the highly restricted architectures ended up in a very conservative or ‘paranoid’ state. These models learned to reject almost all presentations as spoofing attempts in order to artificially minimize their training loss. While this is technically a perfect defense against presentation attacks, it makes the biometric system practically unusable because of the huge rejection of genuine users. This shows that they were not able to learn real material textures that would fail under the extreme lighting of Protocol 4, due to their limited parameter capacities.

In contrast, EfficientNet-B0 was able to break this paranoid cycle and show a robust and balanced performance. As shown in Table III, the evaluation metrics for EfficientNet-B0 illustrate that the APCER was maintained at a very low rate of approximately 3%, while the BPCER was significantly reduced to around 27% thus resulting in a greatly improved ACER of approximately 15%.

The benchmarking results showed that the best overall performance for the presentation attack detection task on the OULU-NPU benchmark was achieved by EfficientNet-B0. While other lightweight architectures suffered a large performance degradation in the challenging situations of Protocol 4, EfficientNet-B0 still showed relatively good generalization ability. The model achieved the lowest ACER (15%) across all evaluated architectures, with an APCER of 3% and a BPCER of 27%. These results show that EfficientNet-B0 provided a better balance between the performance of attack detection and bona fide user acceptance, thus being the best candidate for practical deployment on edge devices.

C. Model Selection and Computational Trade-off

Proper evaluation of the trade-off between computational complexity and biometric accuracy is required to select the ideal deep learning architecture. A model that is too lightweight but inaccurate is not good in practice, and a model that is very accurate but computationally intensive cannot support real-time performance on typical consumer hardware. The combined structural complexity and the final biometric error rates of the three architectures are summarized in Table III to find the optimal baseline for the proposed two-stage authentication pipeline.

TABLE III. Performance and Complexity Comparison of the Evaluated Architectures

Architecture	Total Param.	Trainable Param.	Final ACER
ShuffleNetV2	1,274,430	20,826	50%
MobileNetV2	2,244,954	21,082	$\pm 45\%$
EfficientNet-B0	4,028,630	21,082	$\pm 15\%$

The risk of aggressively reducing the number of model parameters is clearly illustrated by the empirical results in Table III. The most efficient architecture, ShuffleNetV2, with only 1.27M parameters, totally failed to perform the pixel-wise extraction task, resulting in unusable 50% ACER due to absolute mode collapse. Similarly, MobileNetV2 failed to generalize in the uncontrolled lighting environment of Protocol 4, resulting in an unacceptable BPCER that compromised the usability of the system. These results suggest that there is a minimum level of architectural capacity needed for a neural network to be able to parse microscopic discrepancies in material and texture in pixel-wise supervision.

Therefore, the EfficientNet-B0 was selected as the champion model for the integrated system architecture. Among the three lightweight alternatives, it has the highest total parameter count (4.02 million parameters), but it is the only network that achieved balanced and statistically valid biometric performance, with an ACER of about 15%.

This translates into a lightweight model that requires little computational memory, which is crucial from an edge-computing perspective as a footprint of 4.02 million

parameters remains remarkably compact. This ability enables the network to process real-time webcam video frames with ease, even without the need for high-end dedicated GPU hardware. Therefore, EfficientNet-B0 represents the most effective sweet spot, providing the required representational capacity to solve the system’s “paranoid” behaviour, while completely respecting the lightweight constraints for enabling smooth real-time authentication.

D. Real-Time Two-Stage Authentication performance

In order to assess the system’s potential for practical usage, the selected liveness detector EfficientNet-B0 and ArcFace recognition module were deployed into a real-time webcam application. The pipeline was tested in normal indoor lighting conditions, simulating a typical edge-device authentication scenario such as a laptop webcam login.

The system runs in a sequential two-stage pipeline to make the best use of the computational resources. When the user puts their face in front of the camera, the system first detects the facial bounding box. The region of interest must pass the EfficientNet-B0 liveness gate before any facial embeddings for identity recognition are calculated. If the input is classified as a “Spoof”, the authentication attempt is blocked immediately. In addition, a frame-skipping optimization (processing every third frame) was implemented. This mechanism limits the intensive neural network inferences, offering a stable and seamless video frame rate, without computational lag on standard edge hardware.

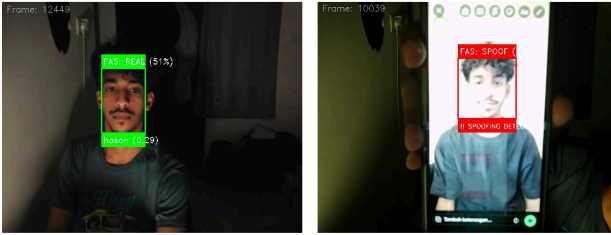


Fig. 5. Implementation results demonstrating system inference on a genuine face (left) and a spoof attack (right).

Figure 8 shows that the integrated system is able to distinguish real presentation from spoofing presentation attacks in real-time. The left panel shows the behavior of the system when processing an authorized user. First, the robust InsightFace module successfully extracts the face bounding box. Then the EfficientNet-B0 liveness detector outputs a spatial pixel-wise map for the cropped region. If the input is classified as a “REAL” presentation, the pipeline proceeds to the second stage. Then, the ArcFace module calculates the facial embedding and compares it with the local database and successfully verifies the person’s identity (e.g., “Hasan”) as shown by the green bounding box.

The right panel shows the defense mechanism of the system during a presentation attack. Once a digital photo is presented to the screen of a smartphone, the frontline liveness detector immediately detects the material anomaly and classifies the input as “SPOOF”. Thus, the system rejects the authentication attempt, shows a red warning interface and bypasses the computationally intensive ArcFace module.

A major advantage of the Deep Pixel-wise Binary Supervision method is the interpretability of its predictions. The spatial prediction maps (heatmaps) produced by the liveness detector were extracted and visualized in order to better understand the model’s decision making.

Unlike the usual global classifiers, the pixel-wise model outputs a 14x14 spatial map scoring each local patch of the input image. In these visualizations, the color spectrum shows the confidence of the model in the spatial predictions: warmer colors (e.g., red and yellow) indicate predicted values close to 1, strongly showing the detection of spoofing artifacts such as digital moiré patterns, flat paper textures, or screen glare. The cooler colors (like green and dark blue) on the other hand are closer to 0, which means that there are no anomalies and that there is a real human skin.

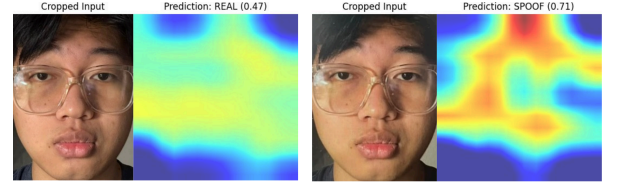


Fig. 6. Visualization of the pixel-wise prediction heatmaps generated by the EfficientNet-B0 model during inference on spoof attack.

The system offers explainability, i.e., visual evidence of the process of decision making, as shown in a side-by-side heatmap comparison in Figure 9. The left panel shows the spatial map obtained with a real presentation. In this case, the heatmap shows a rather uniformly distributed feature with predominantly cool colors, meaning there are no major material anomalies in the face and the prediction is “REAL”.

On the other hand the right panel shows the framework’s response to a presentation attack. This spatial map is heavily marked with spoofing artifacts by a distinct presence of warm red and yellow regions. The high density of warm spots, in turn, visually confirms that the model is successful at localizing anomalous materials like screen glare or digital pixelation, not just analyzing the global facial geometry. The accumulation of these high-scoring pixels eventually leads the global prediction to “SPOOF”, demonstrating the interpretability of the architecture and its robust defense against extreme environmental variations.

V. CONCLUSION

In this study, a two-stage biometric authentication system is proposed using Deep Pixel-wise Binary Supervision (DeepPixBis) for face anti-forgery and face recognition with ArcFace. The proposed method aims at improving robustness to presentation attacks in difficult lighting conditions, while keeping computational efficiency for deployment on edge devices.

To evaluate the effectiveness of the proposed framework, we tested three lightweight CNN backbones ShuffleNetV2, MobileNetV2 and EfficientNet-B0 with Protocol 4 of the OULU-NPU dataset. Experiments demonstrate that DeepPixBis can train the model to learn local texture representations associated with forgery under pixel supervision. The best overall performance among the evaluated architectures was achieved by the EfficientNet-B0 with an APCER of approximately 3%, a BPCER of 27%, and an ACER of 15%. However, both MobileNetV2 and

ShuffleNetV2 have a serious performance degradation under extreme lighting variations, which means that the sufficient representation capacity is demanded for the pixel-by-pixel facial anti-spoofing.

Moreover, the practical applicability of the selected model in resource-constrained environments was illustrated by implementing it into a real-time authentication pipeline using a webcam. The results show that DeepPixBis is an effective and interpretable framework for anti-fake face recognition, focusing on local texture inconsistencies instead of relying only on global facial features.

Despite these promising results, the remaining BPCER shows that the system still has some limitations in terms of handling some real samples under highly uncontrolled lighting conditions. Future work may involve the integration of temporal features, e.g., eye blink and face motion analysis, and the use of multimodal sensing technologies such as depth and infrared imaging to further improve robustness and generalization.

VI. DATA AVAILABILITY

This study utilizes the OULU-NPU dataset. Official access requires submitting an End-User License Agreement (EULA) to the OULU administrators, which can be completed through the official download portal at: <https://sites.google.com/site/oulunpudatabase/download?autouser=0>

In addition, a public version of the dataset is available on Kaggle for testing purposes, which can be accessed at: <https://www.kaggle.com/datasets/mizaku/oulunpu-test>

VII. AUTHOR CONTRIBUTORSHIP

Edwin Antonie provided technical oversight during the study process and led the writing of the methodology, implementation, results, and discussion sections. **Hasan** handled the software investigation and contributed to writing the methodology. **David Christian Golden Mahaviro** contributed to the conceptualization, drafted the introduction and related works, and offered criticism on the manuscript's composition, editing, and review. **Arya Krisna Putra** and **Irene Anindaputri Iswanto** contributed through validation and supervision of the research.

VIII. AI USAGE DECLARATION

We declare that we utilized AI tools, such as Gemini by Google and Claude by Anthropic, during the preparation of this manuscript for grammar checking, translation, and paraphrasing suggestions. All AI-assisted outputs were carefully reviewed, edited, and verified by the authors. Therefore, the authors take full responsibility for the final content and integrity of this paper.

REFERENCES

- [1] Z. Boulkenafet, J. Komulainen, L. Li, X. Feng, and A. Hadid, "OULU-NPU: A Mobile Face Presentation Attack Database with Real-World Variations" *12th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2017)*, May 2017, doi: 10.1109/FG.2017.77
- [2] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A Unified Embedding for Face Recognition and Clustering" *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2015, doi: 10.1109/CVPR.2015.7298682
- [3] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks" *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2018, doi: 10.1109/CVPR.2018.00474
- [4] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive Angular Margin Loss for Deep Face Recognition" in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2019, doi: 10.1109/cvpr.2019.00482
- [5] D. Wen, H. Han, and A. K. Jain, "Face Spoof Detection With Image Distortion Analysis" *IEEE Transactions on Information Forensics and Security*, vol. 10, no. 4, Apr. 2015, doi: 10.1109/TIFS.2015.2400395
- [6] Y. Atoum, Y. Liu, A. Jourabloo, and X. Liu, "Face anti-spoofing using patch and depth-based CNNs" *IEEE International Joint Conference on Biometrics (IJCB)*, Oct. 2017, doi: 10.1109/btas.2017.8272713
- [7] Y. Liu, A. Jourabloo, and X. Liu, "Learning Deep Models for Face Anti-Spoofing: Binary or Auxiliary Supervision" in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2018, doi: 10.1109/cvpr.2018.00048
- [8] Z. Yu, C. Zhao, Z. Wang, Y. Qin, Z. Su, X. Li, F. Zhou, and G. Zhao, "Face Anti-Spoofing with 3D Central Difference Convolutional Networks" *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2020, doi: 10.1109/cvpr42600.2020.00534
- [9] A. George and S. Marcel, "Deep Pixel-wise Binary Supervision for Face Anti-Spoofing" *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 3, no. 1, Jan. 2021, doi: 10.1109/icb45273.2019.8987370
- [10] S. Liu, P. C. Yuen, S. Zhang, and G. Zhao, "3D Mask Face Anti-Spoofing with Remote Photoplethysmography" *European Conference on Computer Vision (ECCV)*, Sep. 2016, doi: 10.1007/978-3-319-46478-7_6
- [11] R. Shao, X. Lan, J. Li, and P. C. Yuen, "Multi-adversarial Discriminative Deep Domain Generalization for Face Anti-spoofing Attack Detection" *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2019, doi: 10.1109/cvpr.2019.01026
- [12] Y. Zhang, Z. Yin, Y. Li, G. Yin, J. Yan, J. Shao, and Z. Liu, "CelebA-Spoof: Large-Scale Face Anti-Spoofing Dataset with Rich Annotations" *European Conference on Computer Vision (ECCV)*, Aug. 2020, doi: 10.1007/978-3-030-58610-2_5
- [13] Z. Yu, C. Zhao, Z. Wang, Y. Qin, Z. Su, X. Li, F. Zhou, and G. Zhao, "Searching Central Difference Convolutional Networks for Face Anti-Spoofing" *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2020, doi: 10.1109/cvpr42600.2020.00534
- [14] K. Patel, H. Han, and A. K. Jain, "Secure Face Unlock: Spoof Detection on Smartphones" *IEEE Transactions on Information Forensics and Security*, vol. 11, no. 10, Oct. 2016, doi: 10.1109/tifs.2016.2578288
- [15] G. Pan, L. Sun, Z. Wu, and S. Lao, "Eyeblink-based Anti-Spoofing in Face Recognition from a Generic Webcam" *11th IEEE International Conference on Computer Vision*, Oct. 2007, doi: 10.1109/icc.2007.4409068
- [16] S. Fatemifar, S. R. Arashloo, M. Awais, and J. Kittler, "Client-Specific Anomaly Detection for Face Presentation Attack Detection" *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 3, no. 4, Oct. 2021, doi: 10.1016/j.patcog.2020.107696
- [17] J. Stehouwer, H. Han, F. Lio, and A. K. Jain, "On the Detection of Digital Face Manipulation" *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2020, doi: 10.1109/cvpr42600.2020.00582
- [18] X. Tu, Z. Ma, J. Zhao, G. Du, M. Xie, and J. Feng, "Learning Generalizable Face Anti-Spoofing via Multi-Domain Network" *Proceedings of the European Conference on Computer Vision (ECCV)*, Aug. 2020, doi: 10.1145/3402446
- [19] H. Chen, Y. Chen, X. Tian, and R. Jiang, "A Cascade Face Spoofing Detector Based on Face Anti-Spoofing R-CNN and Improved Retinex LBP" *IEEE Access*, vol. 9, Feb. 2021, doi: 10.1109/access.2019.2955383
- [20] T. de Freitas Pereira, A. Anjos, and S. Marcel, "Can face anti-spoofing countermeasures work in a real world scenario?"

Information Fusion, vol. 80, Apr. 2022, doi: 10.1109/icb.2013.6612981

- [21] Y. Liu, J. Stehouwer, A. Jourabloo, and X. Liu, "Deep Tree Learning for Zero-shot Face Anti-spoofing" *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2019, doi: 10.1109/cvpr.2019.00481
- [22] L. M. Li, B. W. Wang, X. Wang, P. K. Wang, Y. D. Zhang, and Y. Wang, "Face Anti-Spoofing with Unknown Attacks: A Comprehensive Feature Extraction and Representation Perspective" *Journal of Computer Science and Technology*, vol. 39, no. 4, Jul. 2024, doi: 10.1007/s11390-024-4164-7
- [23] A. George, Z. Mostaani, D. Geissenbuhler, O. Nikisins, A. Anjos, and S. Marcel, "Biometric Face Presentation Attack Detection With Multi-Channel Convolutional Neural Network" *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 42-55, 2020, doi: 10.1109/tifs.2019.2916652
- [24] Y. Liu, J. Stehouwer, and X. Liu, "On Disentangling Spoof Trace for Generic Face Anti-spoofing" *Proceedings of the European Conference on Computer Vision (ECCV)*, Aug. 2020, doi: 10.1007/978-3-030-58523-5_2
- [25] S. Zhang, X. Wang, J. Liu, C. Tang, T. Chen, C. Hou, A. Li, X. Yin, and L. Luo, "CASIA-SURF: A Large-Scale Multi-Modal Benchmark for Face Anti-Spoofing" *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 2, no. 2, Apr. 2020, doi: 10.1109/tbiom.2020.2973001
- [26] H. Li, P. He, S. Wang, A. Rocha, X. Jiang, and A. C. Kot, "Learning Generalized Deep Feature Representation for Face Anti-Spoofing" *IEEE Transactions on Information Forensics and Security*, vol. 13, no. 10, Oct. 2018, doi: 10.1109/tifs.2018.2825949
- [27] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An Extremely Efficient Convolutional Neural Network for Mobile Devices" *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2018, doi: 10.1109/cvpr.2018.00716
- [28] J. Hamblin, K. Nikhal, and B. S. Riggan, "Understanding Cross Domain Presentation Attack Detection for Visible Face Recognition" *16th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2021)*, Dec. 2021, doi: 10.1109/fg52635.2021.9667008
- [29] Y. Xu, Z. Wang, Hu. H, L. Wu, and Y. Lui, "Exploiting Non-uniform Inherent Cues to Improve Presentation Attack Detection" *IEEE International Conference on Multimedia and Expo (ICME)*, Aug. 2021, doi: 10.1109/ijcb52358.2021.9484389
- [30] U. Muhammad, M. Oussalah, "Self-Supervised Face Presentation Attack Detection with Dynamic Grayscale Snippets" *17th IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, Jan. 2023, doi: 10.1109/fg57933.2023.10042547